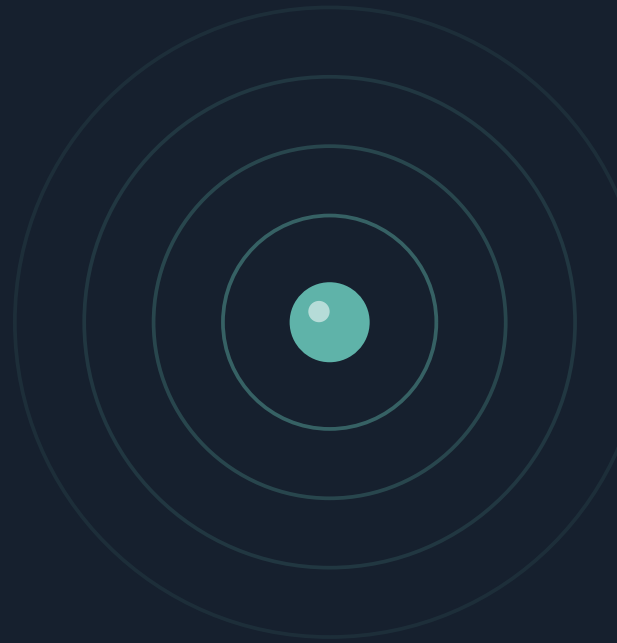




SESSION SUMMARY

Building AI Voice Agents

How to prompt, set up, connect and test an inbound voice receptionist with Retell AI, Cal.com and n8n.

SESSION DATE

September 19, 2026

PROGRAM

Day 6, first session

PLATFORM

Retell AI

MENTOR

Matthew Cohn

HOST

Sumedha

FORMAT

Teaching, then workbook

Contents

Session at a Glance	2
Opening and Housekeeping	3
What a Voice Agent Is	4
Why Voice AI Matters	4
How a Voice Agent Works	5
Prompting: Context and Identity	5
Functions and APIs	6
The Use Case: Bright Smile Dental	7
Retell AI: Platform Tour	7
Creating an Agent	10
Agent Settings That Matter	10
The Prompt, Section by Section	14
Setting Up Functions With Cal.com	17
Post-Call Properties	18
Webhooks and the End-of-Call Report	19
The Live Test Call	19
n8n: Writing the Call Log to Google Sheets	20
Resources and the Bonus Prompt Builder	21
VoiceLink and Phone Numbers for Learners in India	22
Workbook and Close	22
Common Mistakes and Misconceptions	23
Action Steps	23
Key Takeaways	24
Glossary	25

Session at a Glance

This was the first session of Day 6, the first full-day schedule in the program. The topic was AI voice agents. The mentor was Matthew Cohn, founder of Future Flow AI, an agency in Scotland that builds voice and text agents for businesses. The teaching part ran a little over 3 hours. A breakout workbook followed.

The first half covered the basics and a tour of Retell AI, the platform Matthew uses for all client work. The second half broke down a finished agent: a receptionist called Emma for a made-up dental practice, Bright Smile Dental. Emma answers calls, collects details, checks a calendar, books an appointment and logs the call to a Google Sheet through n8n. Matthew called the agent live on screen and the booking landed in his Google Calendar.

The session had one main idea. A voice agent is only as good as its prompt. Every prompt does 2 things: it gives the agent context and identity.

What the Session Covered

- What a voice agent is, the common use cases and the types.
- Why voice AI is a strong market.
- How a voice agent works, step by step.
- Prompting with context and identity.
- Functions and APIs, which let an agent take actions during a call.
- A tour of the Retell AI platform and the settings that matter for beginners.
- A full prompt breakdown: 5 core sections plus supporting sections.
- Calendar functions through Cal.com.
- Post-call properties, webhooks and the end-of-call report.
- A live test call, then a fix found from that test.
- An n8n workflow that writes each call into a Google Sheet.
- A bonus document that builds a voice agent prompt for any use case.
- VoiceLink, a way for learners in India to get a virtual phone number.

What Learners Walked Away With

- The full agent prompt.
- A file to import the finished agent into Retell AI with all settings.
- The post-call property descriptions in a document.
- The Google Sheet template.
- The n8n workflow.
- The slide deck.
- The bonus discovery form and prompt builder.
- A Twilio import guide and the VoiceLink coupon and video.

THE MAIN MESSAGE

Spend your early time on prompting and testing. Build agents, break them, read the transcripts and fix the prompt. That is the difference between a good agent and a great one.

Opening and Housekeeping

Plan for the Day

Sumedha hosted. The session started about 10 minutes late to let people join. She thanked those joining early or at odd hours, since the previous day had ended late. All times below were given as approximate and are in IST.

Time	What happens
10:00 AM to 1:00 PM	Session 1 learning: voice agents.
1:00 PM to 3:00 PM	Voice agent workbook in breakout rooms.
3:00 PM to 4:00 PM	Break.
4:00 PM to 6:30 PM	Session 2 learning: vibe coding and building an AI MVP.
6:30 PM to 7:30 PM	Breakout rooms for questions. No workbook. Questions go into Slido.

- She suggested a working lunch, since the break comes late.
- Recordings, transcripts, resources and the Q and A dump are shared for both sessions.
- She asked learners not to sit back and watch. Keep systems ready and watch closely, because the breakout uses the same kind of build.

Why This Topic Now

For 5 days the group worked with AI in text: prompts, workflows and automations. This session adds voice. The goal was not only to connect apps and make an agent. It was to see which use cases fit and what each learner could do with the skill.

About the Mentor

- Matthew Cohn, from Scotland. It was 5:45 AM for him when he started.
- Founder of Future Flow AI, an AI voice and text agent agency.
- The agency helps businesses capture, qualify and convert more leads. That is all it does.
- About 2 years in voice AI. He has taught Outskill learners before.
- Shares free tips on LinkedIn and YouTube. LinkedIn is his main channel.
- He had a cold and apologized for it more than once.

How He Ran the Session

- 359 people were on the call, from Aberdeen, Amsterdam, Melbourne, Cape Town, the Philippines, Saudi Arabia and more.
- He asked for short replies in the chat to keep energy up.
- Long questions were held for the end.
- He told learners not to take notes or screenshots. Everything is in the resource pack.
- He said he tends to go too deep into the platform. So he showed only what a beginner needs now.
- He teaches only what he uses every day. Where he had not used a feature, he said so.

What a Voice Agent Is

His definition: a smart assistant that understands human speech, uses AI to process the request, and replies in a natural, human-sounding voice. Siri on the iPhone is the simplest example.

The 3 Most Common Uses

- **Customer service agent.** Answers support calls, answers questions and raises tickets.
- **Outbound sales agent.** Calls leads, does discovery, qualifies them and books appointments.
- **Inbound receptionist.** Answers the phone for a business, answers questions and books appointments. He called this the most common use.

Types of Voice Agent

- **Inbound.** Answers calls.
- **Outbound.** Makes calls.
- **Interactive learning assistant.** A voice agent that lives inside an app, website or platform and talks to people while they use it.

Why Voice AI Matters

He admitted he is biased after 2 years in the field. Then he gave the numbers he uses.

Figure	What it means
\$100 billion a year	Business lost by UK and US companies combined, only because calls go unanswered.
21 times	Leads contacted within 5 minutes are 21 times more likely to convert than leads contacted within 30 minutes. This is called speed to lead.
\$35 billion by 2033	Projected size of the AI voice agent market, growing close to 40% a year. He put today's market at about \$5 billion to \$7 billion.

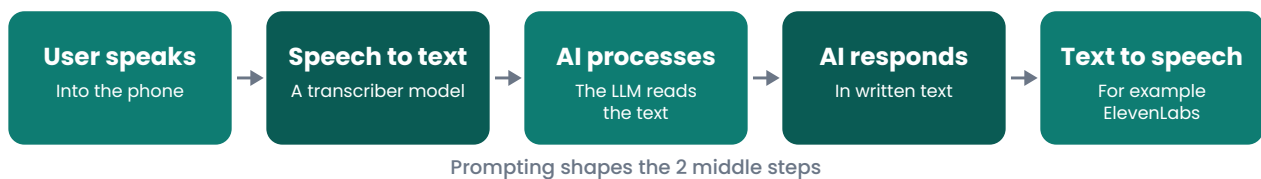
Big Problems, Low Cost

Voice agents solve large problems at a very low cost. They answer every call. They can call every lead within 5 minutes. They work 24 hours a day, 7 days a week. They take no sick days, no holidays and no breaks. A learner added that they never ask for a raise. Matthew called this every business owner's dream employee.

His conclusion: demand will rise sharply. It pays to learn the skill now, for your own business or to sell to other businesses, as he does.

How a Voice Agent Works

Under the surface, a call is a chain of 5 steps. The AI never hears sound. It reads text and writes text.



- The caller speaks.
- A transcriber model turns the speech into text.
- The text goes to an AI model, which processes it.
- The AI writes its reply as text. This step and the one before are where the prompt does its work.
- A text-to-speech model, such as ElevenLabs, turns the reply into a voice.

Prompting: Context and Identity

Prompts are how a voice agent is created. He asked learners to remember 2 words and tested them on it later: context and identity. A prompt tells the agent 3 things.

- Who it is supposed to be.
- What it is supposed to be doing.
- How it is supposed to speak.

He stressed that prompting is not code. It is plain words, instructions and sentences.

THE RULE

Your voice agent is only as good as your prompt. A bad prompt gives a bad agent. Spend a lot of time early on writing prompts, testing them and changing them to see what works.

Bad Prompt and Better Prompt

Version	Prompt	What it gives the agent
Weak	You are a bot that answers questions.	Almost no context. Almost no identity.
Better	You are a friendly assistant that helps customers track orders and gives clear solutions.	Identity: friendly, clear. Context: helps customers, tracks orders.

The better version adds only 7 or 8 words. Those words carry 4 extra pieces of context and identity.

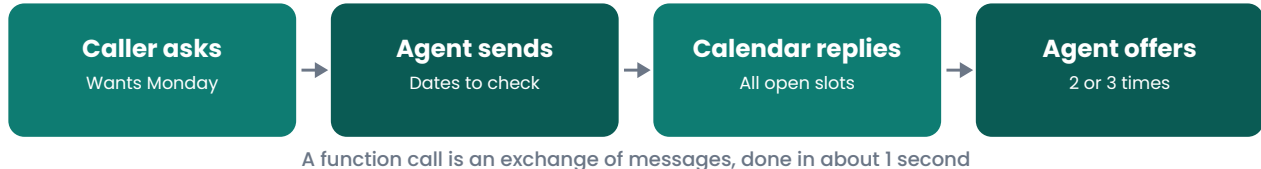
Talking Is Not Enough

An agent that talks on the phone is a nice start. An agent that does tasks the way a person would is where it gets useful. The models are smart, but they need help to do real things, such as checking an order in a database or booking a slot in a calendar. That help comes from functions.

Functions and APIs

What a Function Is

A function lets the agent send and receive information while it is still on the call. A function call takes about 1 second. When an agent says it will check the calendar, that is a function call running.



The Calendar Example

- The agent can send one date and time and ask if it is free.
- More often it sends a day, or 2 to 3 days, and asks for every open slot.
- Example: the caller asks for Monday. The agent asks the calendar for Monday to Wednesday, in case Monday does not work.
- The calendar returns all open slots. The agent reads them and offers a few.
- If the caller then asks for 11:00 AM, the agent already has the list. It can answer at once, or offer other times.

APIs

Most of the group already knew APIs and MCPs. His plain version: an API, or application programming interface, lets you read from or write to a piece of software without having it open in front of you. APIs are what make a voice agent useful to a business. Without them, the agent can talk but gives little value.

- **CRM.** The agent answers calls, takes details and books appointments. All of it is saved in the customer system.
- **Database.** A support agent for an online store looks up an order by name or zip code, then helps with a refund or a status question.
- **Website, app or software.** This is how interactive learning assistants are built.

The Use Case: Bright Smile Dental

The Problem

The normal working day is 9:00 AM to 5:00 PM. Customers also call before and after those hours. Most businesses do not answer then. He tied this back to the \$100 billion figure. Much of that loss is after hours.

- **Missed hot leads.** These people called the business asking for its service.
- **Wasted marketing spend.** Ads, SEO, billboards and signs drive calls that nobody answers.
- **Leads lost to competitors.** He said this one would annoy him the most.

The Offer

Promise a business owner that they will never miss a call, at any day or time, without hiring more staff. The solution is an inbound voice agent that answers every call, gathers details, speaks in a human way, books the appointment and updates the records.

What Was Built

- A made-up dental practice, chosen because everyone can relate to it.
- An inbound agent named Emma.
- She collects the caller's details.
- She books the appointment in the business calendar.
- Each call is written to a Google Sheet as a call log.

A real business may use a CRM in place of a sheet. He left CRMs out to keep the first build simple. He said learners who continue will need to learn CRM integration, and that the Google Sheet is a good first step toward it.

Retell AI: Platform Tour

Retell AI is where the agent is built. Matthew has used it for about 2 years, in his agency and for every client.

Pricing and Sign-Up

- Free to sign up with an email and phone number.
- New accounts get trial credits. He thought it was about \$10.
- No subscription. You pay only for what you use.

- His account showed a long list of agents because it is his testing and demo account.

The Sidebar, From the Bottom Up

Area	What it does	His advice for beginners
Settings	Workspace name and users. Invite team members here.	Simple. Little to cover.
Webhooks (account level)	Account-wide webhook settings.	Ignore. He has never touched it.
API keys	Used when an outside tool such as n8n tells Retell to start a call.	Only needed for outbound agents. Know where it is.
Reliability and limits	Platform limits.	Ignore.
Billing	Billing history and monthly charges.	Self-explanatory.
Integrations	New. Ready-made links to CRMs such as HubSpot, Salesforce and GoHighLevel.	Useful later. Not covered.
Monitor	Analytics, live monitoring, quality checks and alerts.	Skip until an agent is live with real customers.
Contacts	A place to store names, numbers and emails.	He does not use it. Most builders already have a CRM.
Call and chat history	A record of every conversation.	Important. Used to audit agents.
Batch call	Calls a large list of people in one go.	Skip. He called it pointless.
Phone numbers	Buy or connect numbers and link them to agents.	Needed for real calls.
Knowledge bases	Large reference documents for agents.	Important. Covered below.
Agents	Where agents are created and edited.	The core of the platform.

On Monitor, he gave his reason for skipping it. A beginner's only goal is to learn to build very good agents. Monitoring matters once an agent is in production. There are many videos and Retell documents on it, and they make more sense after the basics.

Call History

Retell also supports chat and text agents, so there is a chat history too. His was empty. Call history can be filtered by agent. He noted that Retell holds GDPR, HIPAA and SOC 2 certification. Opening one call shows:

- The phone numbers and the call length.
- The cost. His example call cost 41 cents.

- LLM token use.
- The recording, which can be played or downloaded.
- The conversation analysis, the summary and the transcript.

Preset Analysis

- Whether the call was successful.
- Call status. In history this always reads ended, so he finds it of little use.
- User sentiment. Was the caller happy or aggressive.
- Disconnection reason. For example, the user hung up.
- End-to-end latency. The agent's average response time.

Custom Analysis

These are pieces of information you choose to pull out of each call. In the agent builder they are called post-call properties. They are covered in their own section.

LATENCY

Latency is the time between the caller finishing and the agent replying. 2 seconds sounds short. On a phone call it feels very long. Lower latency makes the call feel natural.

Using Transcripts to Audit an Agent

When an agent says something odd, the voice is rarely the cause. The usual cause is a wrong transcription. His example: the caller says I want 10 apples. The transcriber writes I want 10 cars. The agent, built to sell apples, says it does not sell cars. The caller is confused. Reading the transcript shows what the agent actually received. He told learners to ignore the panel on the right of that screen and focus on custom analysis, summary and transcript.

Batch Call

He advised against it. Retell released it just before many countries brought in rules that require consent for AI calls to people. Also, nobody moves a lead list from a CRM into Retell by hand. An automation platform such as n8n does that on its own.

Phone Numbers

- An agent needs a number to answer or make calls.
- Option 1: buy a number inside Retell. Only US and Canada numbers are sold there.
- Option 2: connect your own number through SIP trunking. He has to do this, being in the UK.
 - Buy a number from a telephony provider. He uses Twilio because it is easy to set up.
 - In Twilio, create a SIP trunk. It gives a termination URI, a username and a password.
 - Enter the number and those 3 details in Retell. Add a nickname and save.
 - The nickname has no effect on how the number works.
- He did not show the full steps. Search YouTube for how to connect a Twilio number to Retell AI. It takes 5 to 10 minutes, plus the same again to open a Twilio account.
- Linking a number to an agent is one click. Pick the agent under inbound and choose the latest version.
- One number can carry both an inbound and an outbound agent.

He explained why that last point matters. An outbound agent calls a lead. The lead misses the call and rings back. Without an inbound agent on that number, the call goes nowhere. With one, the lead is captured, qualified and converted.

Knowledge Bases

The group already knew RAG. A knowledge base is the agent's RAG system.

- Example: a business with 20 services. You want the agent to know all 20 in detail.
- Never put all of that in the prompt.
- A longer prompt makes the agent more expensive. It also makes it perform worse.
- Put the services and descriptions in a knowledge base.
- In the prompt, tell the agent that detailed service information is found in the knowledge base.
- A knowledge base still adds tokens, cost and latency. But far less than putting the same text in the prompt.
- Upload it once on the main screen. Then attach it inside each agent that needs it.
- 3 agents for one business can share one knowledge base, so they all give the same answers.

Creating an Agent

Click Create Agent, choose Voice Agent, then choose Single Prompt.

Why Single Prompt and Not Conversational Flow

- He said every agent his agency builds is a single prompt agent.
- **Easier.** Easier to build, edit and audit, because everything is in one place.
- **Better on the call.** The agent sounds better and moves through the conversation better.
- Conversational flow uses a separate prompt for each stage, such as opening, qualification, booking and closing.
- The weakness: people change their minds. A caller at the booking stage may change the product or a detail given earlier.
- The agent is now inside one stage's prompt. It may lack the context or the functions to handle the change.
- He strongly advised beginners to start with single prompt agents.

Agent Settings That Matter

He said this part separates a good agent from a great one. He planned 15 minutes for it and took 25.

Name, Cost, Latency and Tokens

- The agent name has no effect on performance. It is not linked to the prompt.
- The cost per minute is shown and broken down:
 - The LLM, which reads text and writes the reply.
 - The text-to-speech model.

- Retell's voice infrastructure. This is how Retell earns money.
- Background speech removal, at about half a cent.
- His demo agent showed a latency of 730 to 1,000 milliseconds.
- Tokens measure how long the prompt is. The demo agent used 1,400 to 3,500.
 - The system prompt was about 1,000 to 1,300 of those.
 - The function definitions were 67.
 - Conversation history makes up the rest.
- He has built agents of 8,000 to 9,000 tokens. They cost more. Sometimes that is needed.
- The base LLM rate he showed was 1.6 cents a minute for agents under 4,000 tokens.
- Above 4,000 tokens the price rises, and it keeps rising as tokens grow. A 12,000 token agent costs more than a 9,000 token agent.

Model

He used Gemini 3 Flash because it is cheap and fast, and he has tested it with clients. He switched to GPT 4.1 to show the effect. The cost rose to almost 3 times, and latency also went up. Pick a model that is cheap, good and fast.

Voice

- Retell has built-in voices. The choice is small and mostly American.
- Custom providers are supported: MiniMax, Cartesia, Fish Audio, Inworld and ElevenLabs.
- To add an ElevenLabs voice:
 - Open the ElevenLabs voice library and filter, for example by language and accent.
 - Copy the voice ID.
 - In Retell, open custom providers, choose ElevenLabs, add a custom voice and paste the ID.
 - Save. The voice is now on the agent.
- For the demo he picked a Jamaican English voice called Nicole, for fun.
- No single voice fits every use case. His agency reuses some voices, but not one for everything.
- There is no shortcut. Listen to many voices, put one on the agent, press Run Test and talk to it.

Language

- An agent can speak one language or several. Use multi-select for several.
- More languages used to mean much higher latency. Retell has improved this a lot.
- The catch most people miss: the voice must be trained on the language.
- In ElevenLabs, a small number next to a voice, such as plus 10 or plus 12, shows its extra languages.
- One voice he opened covered English, Greek, Korean, Romanian, Hindi, Vietnamese, Arabic, German and more.
- The Nicole voice had no such number. It speaks English only.
- An English-only voice set to other languages will mispronounce words and sound like nonsense. He compared it to reading Italian aloud with no idea what it means.

Welcome Message

- Choose who speaks first: the AI or the user.
- For outbound, the user may speak first, since people answer with hello.
- 9 times out of 10 he chooses AI speaks first. The first speaker gains control of the call.
- He always writes a custom message. He does not let the AI invent the opening line, because it is the first impression.

EMMA'S WELCOME MESSAGE

Hello, you've reached Bright Dental. My name is Emma, and I'm an AI agent, but I promise I'm one of the good ones. How can I help you today?

Pause Before Speaking

This is the gap between the call connecting and the agent's first word. Early on he left it alone, and his agents spoke the instant the call connected. It felt wrong. People answer, lift the phone to the ear, then speak. He sets this to 0.4 to 0.6 seconds.

Functions and Knowledge Base Panels

Functions are set up here. They are covered in a later section. A knowledge base is attached with the Add button. He attached one on screen so learners could watch cost, latency and tokens all go up. So there is a cost and benefit call to make. It is usually worth it.

Speech Settings

- **Background sound.** Options include coffee shop and mountain outdoors. He only uses call center. Callers hear faint talking and typing, and the call feels more real.
- **Response wait time.** How fast the agent replies. Faster is not always better.
 - Agents still struggle to know when a person has finished talking. People stop mid-sentence and end sentences in odd ways.
 - A very fast agent therefore interrupts people.
 - His working range is 700 to 1,500 milliseconds.
 - If latency sits at 500 to 700, slow the agent a little.
- **Interruption sensitivity.** How easily the agent stops when the caller talks over it.
 - Too low, and the agent talks over someone who is trying to correct it.
 - Too high, and it stops at every cough, yes or okay from an active listener.
 - He sets it at about 0.75.
- **Reminder message frequency and pronunciation.** Not needed for now.

Transcription Settings

- Turn on remove noise and remove background speech. It adds a little cost and improves accuracy.
- Set transcription mode to optimize for accuracy. A slightly slower correct answer beats a fast wrong one.
- Leave vocabulary on general.

- Ignore boosted keywords. He said it does not work.

Call Settings

- Voicemail detection, call screening and IVR hang-up matter for outbound agents, not for this build.
 - On voicemail he usually has the agent hang up. It can also leave a message.
 - iPhones and Android phones now screen unknown callers and ask for a name and reason.
 - With call screening on, the agent gives its identity and purpose, such as Emma from Bright Dental calling about your enquiry form. The person sees that and picks up.
- Keypad input detection is not needed.
- **End call on silence.** He sets 1 minute to 1.5 minutes.
- **Max call duration.** Never leave it at the maximum. Billing is per minute, and a few calls left open for 2 hours get costly.
 - His rule: take the average length of this kind of call and double it. A 5 minute average means a 10 minute cap.
- Ring duration is not needed.

Security and Fallback

- Data storage settings control what Retell keeps. It can redact chosen personal details from stored transcripts.
- Retell runs its own private versions of the models, with a guardrail layer on top.
- The guardrails catch people trying to manipulate the agent or push it toward harmful topics.
- He turns on every guardrail for every agent.
- A fallback voice can be set. He finds ElevenLabs very reliable.
- Opt-in secure URLs and default dynamic variables are not needed for now.

Webhook and MCP

The agent-level webhook tells Retell where to send the end-of-call report. It is covered in its own section. Retell also has an MCP setting. He does not use it, so he did not teach it.

His Settings in One Table

Setting	His value
Agent type	Single prompt
Model	Gemini 3 Flash
Who speaks first	AI, with a custom written message
Pause before speaking	0.4 to 0.6 seconds
Target latency	700 to 1,500 milliseconds
Interruption sensitivity	About 0.75
Background sound	Call center, or none
Noise and background speech removal	On
Transcription mode	Optimize for accuracy
End call on silence	1 to 1.5 minutes
Max call duration	2 times the average call length
Safety guardrails	All on

The Prompt, Section by Section

After a 10-minute break, he opened Emma's prompt. He first checked that the group remembered the 2 words. They did: context and identity.

Markdown Formatting

- The prompt uses markdown so the agent can read its structure, the way a person reads a report with headings.
- Hash symbols mark section titles, such as Role, Context and Personality.
- Asterisks make words bold. Bold tells the agent that a word or sentence is important.

The 5 Core Sections

He uses these 5 sections in every voice agent prompt, for clients and for himself. He called them beginner level and also foundation level.

Role

You are Emma, a friendly receptionist at Bright Smile Dental. You help new patients book appointments over the phone. This gives a name, a manner, the business and the people she serves.

Context

You are speaking with someone who has called the practice. They want to book a dental appointment. You make that quick and easy while gathering the details the dentist needs.

Personality

Warm, friendly and professional. A calm, reassuring tone. Efficient but never rushed. Every caller should feel valued. The prompt also says why: many people feel anxious about the dentist. He stressed this. Give the reason behind an instruction, as you would when training a person.

Task

- Welcome the caller warmly.
- Understand the issue or the service they need.
- If it makes sense, collect their details.
- If it makes sense, book the appointment.
- Confirm the details and say what happens next.

DO NOT GIVE THE AGENT ONE SINGLE GOAL

A common mistake is a task section that says only: your goal is to book appointments. An AI with one goal fixes on it. The agent then pushes every caller to book, even when the caller only has a question.

Spread the weight over the whole conversation. 4 to 6 bullet points is ideal. Do not list 10 or 20.

Conversation Stages

He calls this being a conversational architect. You design the call before it happens. You teach the agent how these calls usually go and how to act at each point. Keep it general and logical. Do not try to map every possible path.

Stage	What the prompt tells Emma
Greeting	Open warmly, introduce yourself and the practice, ask how you can help. The prompt notes that the welcome message already does this. It is there so the agent knows where it is in the call.
Identify the issue	Listen to why they called. Ask about the dental concern or service. Show empathy if they mention pain or discomfort.
Collect details	Ask for the full name, then the phone number. One question at a time. Keep replies brief. Then move to the next stage.
Check calendar availability	Ask which day suits them. Run the check availability function. Offer 2 to 3 slots.
Book the appointment	Once the caller confirms a date and time, run the book appointment function. Confirm the details back.
End the call	Thank them, repeat the details, say a confirmation is coming, ask if there is anything else, close warmly.

Getting Names and Numbers Right

- The agent asks the caller to spell their name. Accents, unusual names and background noise all hurt transcription. Spelling raises the odds of a correct name.
- No setting was changed for this. The conversation itself was designed to improve accuracy.
- A wrong name makes a customer feel less valued.
- The agent repeats the name back and must not ask the next question until the caller confirms.
 - Without that line, his agent read the name back and moved straight to the phone number without waiting.
- The same applies to the phone number. Read it back one digit at a time. Do not say 126 as one hundred and twenty-six.
- His point: the AI is very smart and at times lacks basic human sense. Some instructions must be very exact.

Conditional Instructions

Stage 4 introduces what he called conditional formatting. If the caller does this, do that. The rule is to stay broad. Do not write a line for each word a caller might say. That leads to endless conditions and limits the agent.

- If the caller gives several days, ask them to choose one.
- If the caller is unsure, suggest the earliest available time.
- If the requested date and time is free, say so and ask to book it. Do not ask again which time they want.
- If the caller gave only a date and accepts a suggested slot, move to booking.
- If no slot works, ask for another date and run the function again.

On the risk of going round in circles, he said it is rare. The agent picks up whether the caller wants mornings or afternoons. Callers also know a dentist is not open at 9:00 PM or 6:00 AM.

Calling a Function From the Prompt

- The agent must be told exactly when to use a function. Write it into the prompt by name.
- Example: once you have a specific date, use the check availability function for that date.
- Tell it to say something natural first, such as let me just check the schedule for that day.
- This covers the 1 to 2 seconds of silence while the function runs. A human receptionist does the same.

Supporting Sections

- **Services and practice information.** A short list of services and practical details. He put it in the prompt only because it was short. A longer one belongs in a knowledge base.
- **Function usage guidelines.** The stages say when to use a function. This section says what each function does. Calls do not always follow the plan, so the agent must know the tools well enough not to misuse or miss them.
 - Check availability: use it when the caller gives a specific date. Always say you are checking. Offer the 2 to 3 slots closest to the request.
 - Book appointment: use it only after the caller clearly confirms a date and time. Otherwise the calendar fills with duplicate or unconfirmed bookings. Confirm the details after it runs.
- **Important notes.** He adds these only after testing. They are reminders for problems he saw.

- Ask one question at a time.
- Keep replies short and conversational. Nobody wants a long speech on the phone.
- Always repeat back names and phone numbers.
- If the caller mentions pain or an emergency, find the earliest appointment.

Current Date and Time

The models Retell hosts are not connected to the live internet. They do not know today's date. If a caller asks for tomorrow, the agent sends a random date, often about 3 years in the past. The fix is a dynamic variable that inserts the current date and time for a set time zone. It appears at the start and the end of the prompt.

Setting Up Functions With Cal.com

Function Types in Retell

- Built-in: end call, transfer to a human, transfer to another agent, send an SMS.
- Extract dynamic variable and code. Not needed for now.
- Integrations. Very new. He likes it.
- Custom functions.

This build used the Cal.com integration with 2 functions: check availability and book appointment. Cal.com is free to start.

Prepare Cal.com

- Create an event type. He had a title, a description, a duration and an in-person location with the practice address.
- Events can also use Google Meet, Cal video or Zoom links.
- Connect a calendar you own, under Settings, then Calendars. Options include Apple, Outlook, Google and Zoho.
- He removed and re-added his Google Calendar on screen. It took a few clicks.
- Create an API key under Settings, Developer, API keys.
 - He sets keys to never expire, and admitted that is laziness.
 - Rotate keys for security, for example after someone with account access leaves.
 - The key is shown once only. Copy it and store it somewhere safe. It gives full access to the Cal.com account.

Connect It in Retell

- Paste the API key, select the domain and click connect.
- Add check availability. The description, start time, end time and time zone are filled in by the AI.
- The one thing you enter is the event type ID. It is the 7-digit number in the URL of the event type page. It is not in any settings screen.
- A new option sets what the agent says while the function runs. He chose talk while waiting and typed: let me just check the calendar for you.

- Add book appointment the same way, with the same event type ID. Name, phone, time and time zone come from the conversation.
- Bookings now go straight into his Google Calendar.

Post-Call Properties

Post-call properties pull chosen facts out of each call once it ends. Nobody wants to replay every call or read every transcript. His sales example: before a sales call he wants the person's name, what their business does, their problems and their objections, ready to read.

Property	Type	Purpose
Full name	Text	The caller's name.
Reason for call	Text	Why they called.
Service of interest	Text	What they want.
Appointment booked	Boolean	True or false.
Appointment date and time	Text	Filled if they booked.
Reason for not booking	Text	Timing, dislike of talking to an AI, or anything else. He wants to know.
Call outcome	Selector	One of: appointment booked, callback requested, general inquiry, not interested, emergency, hung up, unknown.

WRITE EACH DESCRIPTION LIKE A PROMPT

Most people write one sentence, such as summarize the reason for the call. That works about 70% of the time. He wants 95%, with real detail.

His test: picture a blank ChatGPT window. It has only the transcript and your description. Garbage in, garbage out. Say what you want and why. His descriptions are sometimes very long.

The resource pack has all his descriptions in one document. Give it to Claude or ChatGPT as an example when writing properties for your own agent.

Webhooks and the End-of-Call Report

- After every call, Retell creates an end-of-call report.
- It holds the post-call properties, the summary, the transcript, the date and time, and the recording link.
- You choose where it goes by entering a webhook URL on the agent.
- Many learners did not know what a webhook is. His picture: a letterbox. The postman knows where to put the letters. A webhook is an address that is open to receive information.
- In n8n, add a Webhook node. Set the method to POST and the response to immediately.
- Copy the production URL. Paste it into the agent-level webhook URL field in Retell.
- From then on, every call by that agent lands in that n8n workflow.
- From there it can go to a Google Sheet, a CRM, Airtable or a database.

The Live Test Call

He rang the agent from his own phone and followed the prompt on screen as the call moved through its stages.

- Emma gave the welcome message. He praised her accent and she thanked him, then steered back to his dental needs.
- He said his teeth looked yellow, he felt self-conscious and he had a wedding coming up. He asked about whitening.
- She confirmed they offer it, mentioned the wedding, and asked for his full name, spelled out.
- She read the spelling back and waited for his yes. She did the same with the phone number.
- She missed his answer about the day once and asked him to repeat it.
- He offered Monday, Tuesday or Wednesday. She asked him to pick one. He chose Tuesday.
- She said she would check the schedule, then offered 9:30 AM, 11:00 AM and 12:30 PM on Tuesday, September 22.
- He asked for something later. She said 12:30 PM was the latest and offered to check Wednesday. He took 12:30 PM.
- The booking failed once. She said there was an issue with the phone number and asked for it again.
- He gave it with the country code. She read it back, booked, and confirmed Tuesday, September 22 at 12:30 PM.
- He asked for a joke. Why did the smartphone go to the dentist? Because it had a Bluetooth.
- She closed by repeating the appointment and the practice address.

His Review

- The voice sounded good, and funnier than he expected.
- Latency was right. Not too fast, not too slow.

- The flow felt natural. She brought up the wedding more than once.
- The appointment appeared in his Google Calendar with name, place, description and phone number. He then deleted it.

The Bug and the Fix

APIs accept phone numbers only in international format, known as E.164, with the country code first. He gave his number in local UK format, so the booking was rejected. The function's own description already said the number must be valid E.164. The agent still did not convert it without being told.

LINE HE ADDED TO THE PHONE NUMBER FIELD

If the user gives the phone number in a normal local format, always convert the number into E.164 format.

He saved the agent. His point: you only find these things by testing, and this is how agents are improved.

n8n: Writing the Call Log to Google Sheets

Reading the Report

In n8n, under Executions, the call had arrived and the sheet had been updated. He copied the run into the editor. The report is very large. Most of it is a word-by-word transcript with timing, which nobody needs. The useful parts are near the bottom:

- The from number. On an inbound call this is the caller's own number, so it is captured even if the caller never says it.
- The call analysis block, with the summary and every custom property.
- The recording URL and the full transcript as one field. Use the search box to find them.

Building the Sheet Step

- Create the Google Sheet first, with a column for each item you want.
 - His columns: date of call, name, number, reason for call, service of interest, appointment booked, appointment date and time, call outcome, reason for not booking, call summary, call recording.
- Add a Google Sheets node and choose append row in sheet. Each call gets a new row. 3 calls from one person give 3 rows.
- Connect a Google Sheets credential. It takes about 10 minutes. He pointed to YouTube for the steps.
- Pick the document by ID, by URL, or from a list. The list needs the Google Drive API enabled.
 - He uses the ID. It is the long string in the sheet URL, after d/ and before the next slash.
- Choose Sheet 1. n8n loads every column name.
- With the view set to schema, drag each value from the report into its column.

A Readable Date

Retell sends the call time as a Unix timestamp, which shows in the sheet as a long number. He asked Claude for an n8n expression that gives the current date and time in a readable form. The first answers were not right. A later one worked and gave Saturday, 19th September at 8:55. After a rerun the sheet showed the date and every other field.

Resources and the Bonus Prompt Builder

The Resource Pack

- The post-call properties document.
- The full agent prompt.
- The Google Sheet.
- A JSON file of the agent. Import it into Retell to get his agent with all settings.
- The n8n workflow.
- The slide deck.

The Bonus: Discovery Form Plus Prompt

One sample prompt only goes so far. So he made a document that helps build a prompt for any use case. He joked that it is prompt inception.

- **Part 1: a discovery form.** Questions are highlighted in yellow. He filled it in for Bright Smile Dental to show the level of detail to aim for.
 - Business: name, what it does, the typical customer.
 - Agent: inbound or outbound, the main goal, who it talks to.
 - Conversation: questions to ask and information to collect. This feeds the post-call properties.
 - The flow of the call. This feeds the conversation stages.
 - Personality and set phrases.
 - Frequent questions, concerns and objections.
 - What makes the business different, and its key services.
- **Part 2: a ready prompt for Claude or ChatGPT.** It needs no editing.
- Fill in the form. Download the document as markdown. Open a fresh chat and attach it.
- Tell the AI that the document has 2 parts, a discovery form and a prompt, and ask it to write the voice agent prompt from them.
- He ran it live in a plain Claude chat, with no custom project.
- The output had role, context, personality, task, conversation flow, knowledge base, common questions, handling concerns, function usage guidelines and important notes.
- Ask for the result in markdown inside a code block. Then one copy button gives clean text to paste into a new single prompt agent.

He called this being efficiently lazy. He also says please and thank you to the AI, because manners cost nothing. After pasting the prompt, set the model, voice, speech settings, functions and post-call properties.

VoiceLink and Phone Numbers for Learners in India

After a 5-minute break, Sumedha shared a second resource. It comes from a past Outskill learner.

- A voice agent embedded in a website needs no phone number.
- A receptionist that people can ring needs a virtual phone number.
- Outside India, providers such as Twilio supply these.
- In India, rules on renting numbers make it harder.
- VoiceLink was built by a past learner of the same program. For now it works with Indian numbers only.
- The cohort gets a wallet balance of 600 through a coupon code, to use toward a number and a channel.
- A channel is the number of calls the agent can handle at one moment. She quoted a price of 1.99 for 1 channel. The currency was not clear in the recording.
- The coupon is valid until October 31, 2026.
- Watch the walkthrough video before using the coupon.
- The KYC page changed that morning. An updated video was promised for Monday.
- These are virtual numbers. They are not a SIM or eSIM and are not for personal calls or WhatsApp.

Where to Find It in the LMS

Open Resources and Recordings, scroll to Day 6, and open Resources. It holds the VoiceLink video, the login link, the coupon code with amount and validity, a step-by-step Twilio import guide, and the Drive link with everything Matthew shared.

Workbook and Close

- The workbook use case is a clinic receptionist for an orthopedic clinic.
- It follows the same pattern as the dental build.
- No phone number is connected in the workbook. That is for learners to try later with the guides.
- The workbook link and webinar ID were shared in the chat and the LMS.
- The reflection link comes at the end of the day, because there are 2 sessions.
- Learners who finished could leave for lunch and rejoin with the same link at 4:00 PM IST.
- A breakout room feedback form was shared at the end.
- Sumedha thanked the co-mentors who ran the rooms. The rooms closed at about 3:00 PM IST.

She previewed Session 2: vibe coding with Harshith Vaddiparthi, who would build an AI MVP from scratch in about 3 hours. She also confirmed that the session with Vaibhav is the next day at 10:00 AM IST.

Common Mistakes and Misconceptions

- **A thin prompt.** Little context and identity gives a poor agent.
- **One single task.** The agent becomes pushy. Give 4 to 6 tasks that cover the whole call.
- **Instructions without reasons.** Tell the agent why, as you would a new hire.
- **Large reference text in the prompt.** It costs more and performs worse. Use a knowledge base.
- **Starting with conversational flow.** It breaks when callers change their minds. Start with single prompt.
- **Conditions that are too specific.** Keep if-then rules broad.
- **Not naming the function in the prompt.** Say exactly when to use it, and explain what it does.
- **Booking before the caller confirms.** This creates duplicate or unwanted appointments.
- **No filler line during a function call.** Silence makes callers ask if anyone is there.
- **Not waiting for confirmation.** Tell the agent not to move on until the caller confirms.
- **Chasing the lowest latency.** Too fast means interruptions. Stay between 700 and 1,500 milliseconds.
- **Interruption sensitivity at an extreme.** About 0.75 works.
- **A voice that does not speak the language.** Check the voice's language list.
- **Letting the AI write the opening line.** Write it yourself.
- **No cap on call length.** Open calls cost money.
- **One-line post-call descriptions.** Write them like prompts.
- **Forgetting the date variable.** The agent will book dates years in the past.
- **Local phone number formats.** APIs need E.164.
- **Blaming the voice for odd replies.** Check the transcript first.
- **Losing the Cal.com API key.** It is shown once.
- **Studying analytics too early.** Learn to build good agents first.

Action Steps

- Sign up for Retell AI and use the trial credits.
- Import Matthew's agent file. Read every setting and compare it with the settings table in this summary.
- Create a Cal.com account. Connect your calendar. Create an event type and an API key. Store the key safely.
- Add the check availability and book appointment functions with your event type ID.
- Add the E.164 conversion line and the current date variable.
- Create the post-call properties. Use his descriptions document as the model.
- Build the n8n workflow: Webhook node, then Google Sheets append row. Paste the production URL into the agent.
- Use Run Test and talk to the agent. Try several voices.

- Read the transcript of each test. Add important notes to the prompt for each problem you find.
- Fill in the discovery form for a use case of your own. Generate a prompt with Claude or ChatGPT.
- For real calls, connect a Twilio number through SIP trunking, or use VoiceLink in India after watching the video.
- Later: learn CRM integration, outbound agents and API keys, and the monitoring tools.

Key Takeaways

- A voice agent is speech to text, an AI model, then text to speech. The prompt controls the middle.
- Every prompt gives context and identity: who the agent is, what it does and how it speaks.
- The agent is only as good as the prompt. Test, read transcripts, refine.
- Functions and APIs turn a talking agent into one that does useful work.
- Use 5 core prompt sections: role, context, personality, task and conversation stages.
- Design the conversation itself for accuracy: spell names, read numbers back digit by digit, wait for a yes.
- Settings matter: a cheap fast model, a fitting voice, a short pause, balanced latency and interruption values, and a call length cap.
- Post-call properties and a webhook turn each call into clean data in a sheet or CRM.
- Missed calls and slow follow-up cost businesses a lot. That is the sales case for voice agents.
- The discovery form and prompt builder make it possible to start an agent for any business in minutes.

Glossary

Term	Plain meaning
Voice agent	An AI that understands speech and replies in a human-sounding voice.
Inbound and outbound	Inbound agents answer calls. Outbound agents make them.
Speed to lead	How fast a business contacts a new lead. Faster converts better.
Transcriber	A model that turns speech into text.
LLM	The AI model that reads text and writes the reply.
Text to speech	A model that turns text into a voice.
Latency	The time between the caller finishing and the agent replying.
Tokens	The measure of prompt length. More tokens cost more.
Function	A way for the agent to send and receive information during a call.
API	A way to use a piece of software without opening it.
CRM	A system where a business keeps customer records.
RAG and knowledge base	A store of documents the agent can look things up in.
Single prompt agent	An agent run by one prompt for the whole call.
Conversational flow	An agent type with a separate prompt for each stage.
Markdown	Simple text formatting with hash symbols for titles and asterisks for bold.
Conditional instruction	An if-then rule in the prompt.
Dynamic variable	A value filled in at call time, such as the current date.
Post-call property	A chosen fact pulled from a call after it ends.
Boolean	A true or false value.
Selector	A value picked from a fixed list of choices.
End-of-call report	The data package Retell sends after each call.
Webhook	A URL that is open to receive information, like a letterbox.
SIP trunk	The link that connects a phone number from a provider to Retell.
Termination URI	An address created with the SIP trunk and entered in Retell.

Term	Plain meaning
E.164	International phone number format with the country code first.
Unix timestamp	A date and time stored as one long number.
Event type ID	The number that identifies a Cal.com appointment type.
Channel	In Voicelink, the number of calls handled at one moment.
Production	An agent that is live with real customers.